Quantum Field Theory

Topic 2: Interacting theory

Introduction to Interacting Theory

The Goal: Our objective is to express the interacting vacuum $|\Omega\rangle$ in terms of the free vacuum $|0\rangle$ and interacting field in terms of free fields ϕ_0.

The Roadmap:

1. Define the Kinematics (Lagrangian $\mathcal{L}$ and Action S) and establish the Hamiltonian in Free Theory (H_0).
2. Introduce Interaction Hamiltonian (H_{int}) via a potential $V(\phi_0)$.
3. Use total hamiltonian, H $= (H_0)+(H_{int})$ to derive interacting vacuum and interacting field

Next, we assume familiarity with Feynman propagators $D_F(x-y)$, which serve as the building blocks for these interacting expressions.

VEV in Free Theory

In free theory, the **Vacuum Expectation Value (VEV)** of free field operators allows us to define the propagation of particles in the free vacuum $|0\rangle$.

1. The Wightman Function (Two-Point VEV)

The simplest VEV is the amplitude for a free field at y to affect a free field at x:

$$D(x-y) = \langle 0| \phi_0(x)\phi_0(y) |0\rangle \tag{1}$$

This describes a particle created at y and annihilated at x without any time constraints.

2. Connection to the Feynman Propagator

The Feynman propagator is simply a **time-ordered VEV**:

$$D_F(x-y) = \underbrace{\langle 0| T\{\phi_0(x)\phi_0(y)\} |0\rangle}_{\text{The VEV}} \tag{2}$$

The Feynman Propagator

1. Time-Ordering Definition

$$\boxed{D_F(x-y) = \langle 0| \, T\{\phi_0(x)\phi_0(y)\} \, |0\rangle} \tag{3}$$

- **Time-Ordering (T):** Arranges operators such that the latest time is on the left.
- **Anti-Time-Ordering ($\bar{T}$):** Arranges operators such that the earliest time is on the left.

2. Definition using Step Functions

Using the Heaviside step function $\theta(t)$, we decompose the time-ordered product into its two causal components:

$$\boxed{D_F(x-y) = \theta(x^0 - y^0)\langle 0|\phi_0(x)\phi_0(y)|0\rangle + \theta(y^0 - x^0)\langle 0|\phi_0(y)\phi_0(x)|0\rangle} \tag{4}$$

where $\theta(t) = 1$ for $t > 0$ and 0 for $t < 0$.

The Feynman Propagator

3. Commutator Representation

Using the field decomposition $\phi_0(x) = \phi_0^+(x) + \phi_0^-(x)$, where ϕ_0^+ is the annihilation part and ϕ_0^- is the creation part, the ordinary propagator becomes a commutator:

$$\langle 0|\phi_0(x)\phi_0(y)|0\rangle = \langle 0|[\phi_0^+(x), \phi_0^-(y)]|0\rangle \tag{5}$$

Since $\phi_0^+(x)\,|0\rangle = 0$, only the commutator term survives.

Substituting the commutator expression back into the step-function definition, we obtain the propagator entirely in terms of field components:

$$\boxed{D_F(x-y) = \theta(x^0 - y^0)\,\langle 0|\,[\phi_0^+(x), \phi_0^-(y)]\,|0\rangle + \theta(y^0 - x^0)\,\langle 0|\,[\phi_0^+(y), \phi_0^-(x)]\,|0\rangle} \tag{6}$$

The Feynman Propagator

4. Pole prescription

We begin with the VEV of two free fields $\phi_0(x_1)$ and $\phi_0(x_2)$:

$$\langle 0| \phi_0(x_1)\phi_0(x_2) |0\rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{i\vec{k}(\vec{x}_2-\vec{x}_1)} e^{-i\omega_k(t_1-t_2)} \tag{7}$$

Applying Time-Ordering (T): By definition, T arranges fields so the later time is on the left:

$$\begin{aligned}\langle 0| T\{\phi_0(x_1)\phi_0(x_2)\} |0\rangle = \theta(\tau) \langle 0| \phi_0(x_1)\phi_0(x_2) |0\rangle \\ + \theta(-\tau) \langle 0| \phi_0(x_2)\phi_0(x_1) |0\rangle\end{aligned}$$

After substituting the field expansions and shifting $\vec{k} \to -\vec{k}$ in the first term:

$$= \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{-i\vec{k}(\vec{x}_1-\vec{x}_2)} \left[e^{-i\omega_k\tau}\theta(\tau) + e^{i\omega_k\tau}\theta(-\tau)\right] \tag{8}$$

The Feynman Propagator

To simplify the brackets, we use the identity relating the sum to an ω-integral:

$$\boxed{e^{-i\omega_k\tau}\theta(\tau) + e^{i\omega_k\tau}\theta(-\tau) = \lim_{\epsilon\to 0}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi i}\frac{-2\omega_k}{\omega^2-\omega_k^2+i\epsilon}e^{i\omega\tau}} \tag{9}$$

Derivation via Residues: Using partial fractions:

$$\frac{1}{\omega^2-\omega_k^2+i\epsilon} = \frac{1}{2\omega_k}\left[\frac{1}{\omega-(\omega_k-i\epsilon)} - \frac{1}{\omega-(-\omega_k+i\epsilon)}\right] \tag{10}$$

- **Pole 1 ($\tau < 0$):** $\omega = \omega_k - i\epsilon$. Clockwise contour gives $-2\pi i e^{i\omega_k\tau}\theta(-\tau)$.
- **Pole 2 ($\tau > 0$):** $\omega = -\omega_k + i\epsilon$. Counter-clockwise contour gives $2\pi i e^{-i\omega_k\tau}\theta(\tau)$.

The Feynman Propagator

Now, substitute the identity into the time-ordered integral from Step 1:

$$\langle 0| \, T\{\phi_0(x_1)\phi_0(x_2)\} \, |0\rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{-i\vec{k}\cdot\Delta\vec{x}} \left[\int \frac{d\omega}{2\pi i} \frac{-2\omega_k e^{i\omega\tau}}{\omega^2 - \omega_k^2 + i\epsilon} \right] \tag{11}$$

The Feynman Propagator

$$D_F(x_1, x_2) = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik(x_1 - x_2)} \tag{12}$$

The Free Scalar Field Lagrangian

The dynamics of a free real scalar field $\phi_0(x)$ are described by the **Klein-Gordon Lagrangian density**:

$$\mathcal{L}_0(\phi_0, \partial_\mu \phi_0) = \frac{1}{2}(\partial_\mu \phi_0)(\partial^\mu \phi_0) - \frac{1}{2} m^2 \phi_0^2 \tag{13}$$

$$\mathcal{L}_0 = \frac{1}{2}\dot{\phi}_0^2 - \left[\frac{1}{2}(\nabla \phi_0)^2 + \frac{1}{2} m^2 \phi_0^2\right] \tag{14}$$

Components:

- **Kinetic Term:** $\frac{1}{2}\dot{\phi}_0^2$ represents the energy of field fluctuations in time.
- **Potential Term:** $\frac{1}{2}(\nabla \phi_0)^2 + \frac{1}{2} m^2 \phi_0^2$ represents the energy cost from spatial variations (gradients) and the field mass.
- **"Free" Theory:** It is quadratic in ϕ_0, meaning the resulting equations of motion are linear and easy to solve.

The Action and Equations of Motion

The **Action** S is the spacetime integral of the Lagrangian density:

$$S = \int d^4x \, \mathcal{L}_0 = \int d^4x \left[\frac{1}{2}(\partial_\mu \phi_0)^2 - \frac{1}{2} m^2 \phi_0^2 \right] \tag{15}$$

Principle of Least Action

Varying the action with respect to the field ($\delta S = 0$) yields the **Euler-Lagrange equation**:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_0)} \right) - \frac{\partial \mathcal{L}}{\partial \phi_0} = 0 \tag{16}$$

For our free theory, this results in the **Klein-Gordon Equation**:

$$(\Box + m^2)\phi_0 = 0 \tag{17}$$

The Free Hamiltonian

To find the Hamiltonian, we first define the **conjugate momentum** density:

$$\pi_0(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_0} = \dot{\phi}_0(x) \tag{18}$$

Performing a Legendre transformation, the **Hamiltonian Density** is:

$$\mathcal{H}_0 = \pi_0 \dot{\phi}_0 - \mathcal{L}_0 = \frac{1}{2}\pi_0^2 + \frac{1}{2}(\nabla\phi_0)^2 + \frac{1}{2}m^2\phi_0^2 \tag{19}$$

Total Free Hamiltonian

$$H_0 = \int d^3x \left[\frac{1}{2}\pi_0^2 + \frac{1}{2}(\nabla\phi_0)^2 + \frac{1}{2}m^2\phi_0^2\right] \tag{20}$$

This H_0 is what generates the time evolution of the free vacuum $|0\rangle$.

The Interacting Lagrangian and Potential

To describe particles that scatter, we modify the free Lagrangian by adding an **interaction potential** $V(\phi_0)$:

$$\mathcal{L} = \mathcal{L}_0 - V(\phi_0) = \frac{1}{2}(\partial_\mu \phi_0)^2 - \frac{1}{2}m^2\phi_0^2 - V(\phi_0) \tag{21}$$

Defining the Potentials:

- **Interaction Potential (V):** The new non-linear term, typically **quartic**:

$$V(\phi_0) = \frac{\lambda}{4!}\phi_0^4 \tag{22}$$

- **Field Potential (U):** The sum of the mass and interaction terms:

$$U(\phi_0) = \frac{1}{2}m^2\phi_0^2 + \frac{\lambda}{4!}\phi_0^4 \tag{23}$$

- **Total Potential Energy Density ($\mathcal{V}_{tot}$):** Includes the spatial gradient (elastic energy) and the field potential:

$$\mathcal{V}_{tot} = \frac{1}{2}(\nabla\phi_0)^2 + U(\phi_0) \tag{24}$$

Non-Linearity and Scattering

The addition of the quartic term $\frac{\lambda}{4!}\phi_0^4$ fundamentally changes the behavior of the field equations.

The Interacting Equation of Motion

Applying the Euler-Lagrange equation to the interacting Lagrangian $\mathcal{L}$ yields a **non-linear** partial differential equation:

$$(\Box + m^2)\phi_0 + \frac{\lambda}{3!}\phi_0^3 = 0 \tag{25}$$

Physical Consequences:

- **Breakdown of Superposition:** Unlike the free theory, the sum of two solutions is no longer a solution. The ϕ_0^3 term acts as a "source" where fields interact.
- **Scattering:** This non-linearity allows energy to be exchanged between different modes. When two wave packets (particles) meet, they distort each other and bounce off, resulting in physical scattering.
- **Coupling Strength:** The constant λ dictates how "violently" the particles scatter. If $\lambda \to 0$, we recover the linear free theory.

The Action and Renormalizability

The **Interacting Action** S governs the full dynamics:

$$S = \int d^4x \left[\frac{1}{2}(\partial_\mu \phi_0)^2 - \frac{1}{2}m^2\phi_0^2 - \frac{\lambda}{4!}\phi_0^4 \right] \tag{26}$$

Why the Quartic Term (ϕ_0^4)?

- **Coupling Coefficient (λ):** A dimensionless constant (in 4D) that determines the strength of the interaction.
- **Renormalizability:** In 4D spacetime, terms higher than ϕ_0^4 (like ϕ_0^6) have coupling constants with negative mass dimensions.
- These lead to "non-renormalizable" theories where infinities cannot be controlled. ϕ_0^4 is the highest power that allows for a mathematically consistent, predictive QFT at all energy scales.

The Total Hamiltonian

We perform the Legendre transformation again. Since $V(\phi_0)$ does not contain time derivatives, the conjugate momentum remains $\pi_0 = \dot{\phi}_0$.
The **Total Hamiltonian Density** is:

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{int} = \left[\frac{1}{2}\pi_0^2 + \frac{1}{2}(\nabla\phi_0)^2 + \frac{1}{2}m^2\phi_0^2\right] + \frac{\lambda}{4!}\phi_0^4 \tag{27}$$

The Full Hamiltonian

$$H = H_0 + H_{int} \tag{28}$$

where:

- $H_0 = \int d^3x\, \mathcal{H}_0$ (The Free/Kinetic part)
- $H_{int} = \int d^3x \frac{\lambda}{4!}\phi_0^4$ (The Interaction part)

H is the operator that evolves the system in the presence of scattering.

Goal 1: Expressing ϕ in terms of ϕ_0

We define the time evolution for the two fields starting from $t = 0$:

- **Interacting Field:** $\phi(t, \mathbf{x}) = e^{iHt}\phi(\mathbf{x}, 0)e^{-iHt}$
- **Free Field:** $\phi_0(t, \mathbf{x}) = e^{iH_0 t}\phi_0(\mathbf{x}, 0)e^{-iH_0 t}$

At $t = 0$, we set the fields equal: $\phi(\mathbf{x}, 0) = \phi_0(\mathbf{x}, 0)$. Substituting this into the interacting expression:

$$\phi(t, \mathbf{x}) = e^{iHt}e^{-iH_0 t}\phi_0(t, \mathbf{x})e^{iH_0 t}e^{-iHt} \tag{29}$$

The Time-Evolution Operator $U(t)$

We define the operator $U(t) = e^{iH_0 t}e^{-iHt}$. The relationship becomes:

$$\boxed{\phi(t, \mathbf{x}) = U^{-1}(t)\phi_0(t, \mathbf{x})U(t)} \tag{30}$$

This allows us to calculate interacting field values using known free field operators and a rotation operator $U(t)$.

The Evolution of $U(t)$

To find $U(t)$, we take its time derivative:

$$i\frac{d}{dt}U(t) = i\frac{d}{dt}\left(e^{iH_0 t}e^{-iHt}\right) \tag{31}$$

Using the product rule and the fact that $H = H_0 + H_{int}$:

$$i\frac{d}{dt}U(t) = e^{iH_0 t}(H - H_0)e^{-iHt} = e^{iH_0 t}H_{int}e^{-iHt} \tag{32}$$

The Interaction Picture Hamiltonian

By inserting $I = e^{-iH_0 t}e^{iH_0 t}$, we define $H_I(t) = e^{iH_0 t}H_{int}e^{-iH_0 t}$:

$$\boxed{i\frac{d}{dt}U(t) = H_I(t)U(t)} \tag{33}$$

This is a Schrödinger-like equation where the "Hamiltonian" itself is time-dependent because it is evolved by H_0.

The Time-Ordered Solution

Integrating the differential equation $i\dot{U} = H_I U$ with the initial condition $U(0) = 1$ leads to the Dyson series.

Why Time-Ordering (T)?

The operators at different times do not commute. To maintain causality, we must ensure that operators acting at earlier times are applied first (placed to the right).

The Unitary Evolution Operator

$$\boxed{U(t) = T\left\{\exp\left(-i\int_0^t dt' H_I(t')\right)\right\}} \tag{34}$$

Summary:

- The T symbol instructs us to sort the expanded power series of the exponential by time.
- This expression is the foundation for **Feynman Diagrams** and all perturbative calculations in QFT.

The Dyson Series Expansion

The time-ordered exponential for $U(t)$ is actually a shorthand for an infinite series of nested integrals, known as the **Dyson Series**:

$$U(t) = 1 + (-i)\int_0^t dt_1 H_I(t_1) + \frac{(-i)^2}{2!}\int_0^t dt_1 dt_2 T\{H_I(t_1)H_I(t_2)\} + \dots \quad (35)$$

Physical Interpretation:

- **Order 0:** The field propagates without any interaction (1).
- **Order 1:** Particles meet and interact exactly **once** at time t_1.
- **Order 2:** Particles interact at t_1, then scatter again at t_2.

Why this matters for Perturbation Theory

In most theories (like QED), the coupling constant λ is small. This means:

- The first few terms provide a very accurate approximation.
- Each term in this series corresponds directly to a **Feynman Diagram** of a specific order.

The Bridge to the S-Matrix

The evolution operator $U(t)$ allows us to define the **S-Matrix** (Scattering Matrix), which connects states in the far past ($t \to -\infty$) to the far future ($t \to \infty$):

$$S = U(\infty, -\infty) = T\left\{\exp\left(-i\int_{-\infty}^{\infty} dt\, H_I(t)\right)\right\} \tag{36}$$

From Math to Measurement:

- The **Dyson Series** expands this exponential into a sum of interactions.
- Each term in the series corresponds to a **Feynman Diagram**.
- The S-matrix elements $\langle f|S|i\rangle$ give the probability amplitudes for scattering processes we observe in experiments.

Conclusion: The Interacting Field $\phi(x)$

The interacting field in terms of the free field and the evolution operator

$$\phi(t, \mathbf{x}) = U^{-1}(t)\phi_0(t, \mathbf{x})U(t) \tag{37}$$

Where each component is defined as:

- $\phi(t, \mathbf{x})$: The full interacting field (evolved by H).
- $\phi_0(t, \mathbf{x})$: The free field (evolved by H_0). This is the field we can expand in terms of creation ($a^\dagger$) and annihilation (a) operators.
- $U(t)$: The unitary operator $T\{\exp\left(-i\int H_I dt'\right)\}$. It "rotates" the free field into the interacting basis.

Physical Logic: This structure proves that **all** information about interactions is contained within the operator $U(t)$. If we know the free theory and we can calculate the Dyson series for $U(t)$, we can describe any interacting system.

Goal 2: The Interacting Vacuum $|\Omega\rangle$

We start with the free vacuum $|0\rangle$ and evolve it using the full Hamiltonian H. By inserting a complete set of interacting eigenstates $|n\rangle$:

$$e^{-iHt}|0\rangle = e^{-iHt}\sum_n |n\rangle\langle n|0\rangle = \sum_n e^{-iE_n t}|n\rangle\langle n|0\rangle \tag{38}$$

The Limit Trick: We take the limit $t \to \infty(1 - i\epsilon)$. This small imaginary component ϵ ensures that for all $n > 0$, the exponential $e^{-iE_n t}$ decays to zero, leaving only the ground state $|\Omega\rangle$:

$$|\Omega\rangle = \lim_{t\to\infty(1-i\epsilon)} \frac{e^{-iHt}|0\rangle}{e^{-iE_0 t}\langle\Omega|0\rangle} \tag{39}$$

Physical Interpretation: By evolving the system for an infinite time with a slight "damping" (ϵ), the physics "settles" into its true ground state. The bra $\langle\Omega|$ is derived similarly by evolving to the far past ($t \to -\infty$).

Correlations in the Interacting Vacuum

We want to calculate the correlation $\langle\Omega|T\{\phi(x_1)\phi(x_2)\}|\Omega\rangle$. **Step 1: Substitute the Vacuum Definitions** Using our limit expressions for $\langle\Omega|$ and $|\Omega\rangle$:

$$\frac{\langle 0|e^{-iH(T-t_1)}\phi(x_1)e^{-iH(t_1-t_2)}\phi(x_2)e^{-iH(t_2+T)}|0\rangle}{\langle 0|e^{-iH(2T)}|0\rangle} \tag{40}$$

Step 2: Transition to the Interaction Picture We replace the interacting fields ϕ with $U^{-1}\phi_0 U$. After a long series of cancellations of the $e^{iH_0 t}$ and e^{-iHt} terms:

$$\cdots = \frac{\langle 0|T\{\phi_0(x_1)\phi_0(x_2)U(T,-T)\}|0\rangle}{\langle 0|U(T,-T)|0\rangle} \tag{41}$$

Physical Meaning: The probability of a particle moving between two points in an interacting theory is equal to the free propagation **weighted** by all possible interactions (U) occurring in the background.

Conclusion: The interacting vacuum

Taking the limit $T \to \infty(1 - i\epsilon)$, we arrive at the final normalized expression for the n-point correlation function:

Interacting vacuum in terms of free vacuum

$$\langle\Omega|T\{\phi(x_1)\dots\phi(x_n)\}|\Omega\rangle = \frac{\langle 0|T\{\phi_0(x_1)\dots\phi_0(x_n)e^{-i\int d^4x\mathcal{H}_{int}(\phi_0)}\}|0\rangle}{\langle 0|T\{e^{-i\int d^4x\mathcal{H}_{int}(\phi_0)}\}|0\rangle} \tag{42}$$

Key Components:

- **Numerator:** Generates all Feynman Diagrams (including those where the vacuum "bubbles").
- **Denominator:** Exactly cancels the "Vacuum Bubble" diagrams, ensuring the probability is correctly normalized.
- **Free Vacuum $|0\rangle$:** Everything is now calculated using the simple, solvable free theory.